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SUPERSEDING
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March 12, 1968

FEDERAL SPECIFICATION

GRINDING MACHINES, TOOL AND CUTTER; AND GRINDING AND GASHING MACHINES, TOOL, CUTTER, FLUTE AND RADIUS

This specification was approved by the commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric motor driven tool and cutter; and tool, cutter, flute and radius grinding and gashing machines, bench and floor types, with equipment and accessories as described herein.

1.2 Classification.

1.2.1 Types and sizes. Grinding machines covered by this specification are of the following types, styles, and sizes. The type, style, and size furnished shall be as specified (see 6.2.1):

Type 0 - Tool and cutter grinder, bench or pedestal, manual operation, horizontally mounted wheelhead spindle

1/ Size	Rated tool and cutter diameter, swing over table (inches)	Maximum distance between tailstock centers (inches)
817	8	17

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Type I - Tool and cutter grinder, floor mounted, manual operation

Style A - Horizontally mounted wheelhead spindle.

1/ Basic size	Rated tool and cutter diameter, swing over table (inches)	Maximum distance between tailstock centers (inches)	Optional sizes (same rated tool and cutter diameter, swing over table)	Optional maximum distance between tailstock centers (inches)
815	8	15	822, 824, 832	22, 24, 32
1022	10	22	1027, 1030, 1033	27, 30, 33
1227	12	27	1232	32

Style B - Vertically mounted wheelhead spindle

1/ Size	Rated tool and cutter diameter, swing over table (inches)	Maximum distance between tailstock centers (inches)
815	8	15

Type II - Tool and cutter grinder, floor mounted, power operated, horizontal wheelhead spindle

1/ Basic size	Rated tool and cutter diameter, swing over table (inches)	Maximum distance between tailstock centers (inches)	Optional sizes (same rated tool and cutter diameter, swing over table)	Optional maximum distance between tailstock centers (inches)
815	8	15	817, 822, 824, 832	17, 22, 24, 32
1022	10	22	1027, 1030, 1033	27, 30, 33
1227	12	27	1232	32

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Type III - Tool, cutter, gashing, and radius grinder, floor mounted, horizontally mounted wheelhead spindle

1/ Size	Rated tool and cutter diameter, swing over main slide (inches)	Work length accommodated (inches)
610	6	10

Type IV - Tool, cutter, flute and radius grinder, floor mounted, horizontally mounted wheelhead spindle

1/ Basic size	Rated tool and cutter diameter, swing over main slide (inches)	Work length accom-modated (inches)	Optional sizes (same rated tool and cutter diameter, swing over main slide)	Optional work length accom-modated (inches)
84	8	4	810	10
126	12	6	-	-

1/ When optional or other sizes are required, the machine size and its corresponding characteristics shall be as specified (see 6.6).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-W-17929 - Wheels, Abrasive, Precision.
MIL-M-18058 - Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Occupational Safety and Health Act (OSHA) of 1970:

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards

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National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI), Inc., Standards:

B5.1 - T-Slots - Their Bolts, Nuts, Tongues, and Cutters.
B5.10 - Machine Tapers, Self-Holding and Steep Taper Series.
B74.13 - Markings for Identifying Grinding Wheels.
B7.1 - American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels.
B74.2 - Shapes and Sizes of Grinding Wheels.
C1 - National Electrical Code.

(Application for copies should be addressed to the American National Standards Institute, Department 969, 1430 Broadway, New York, NY 10018.)

National Electrical Manufacturers' Association (NEMA) Standards:

MG-1 - Motors and Generators.
IC-1 - Industrial Controls.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

American Gear Manufacturers Association AGMA:

AGMA 360 - Manual for Machine Tool Gearing.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the product are classified herein as follows:

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Requirement	Paragraph
First Article approval	3.2
General requirements	3.3
Detail requirements, Type 0	3.4
Detail requirements, Type I	3.5
Detail requirements, Type II	3.6
Detail requirements, Type III	3.7
Detail requirements, Type IV	3.8

3.2 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

3.3 General requirements.

3.3.1 Material. Material not definitely specified herein shall be of the kind and quality normally used by the manufacturer for grinding machines of the specified type, style, and size, provided the complete machine shall meet the applicable requirements of Section 3 when tested in accordance with Section 4.

3.3.2 Design. The machine shall be new and one of the manufacturer's current models. The machine shall be capable of resharpening conventional type cutters and producing new cutters when required, to tolerances and finishes in accordance with the requirements herein. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.3.3 Operating controls. The machine shall be completely wired through the controls and ready for connection to the applicable power source. Unless otherwise specified (see 6.2.1), the motor control shall conform to NEMA IC-1 and shall be installed in a NEMA type I enclosure, or better. Unless otherwise specified (see 6.2.1), the machine ON-OFF switch shall incorporate overload protection if not provided in the motor or control. Controls for all functions of the machine shall be arranged in a location convenient to the operator. Controls, such as push buttons, dials, and switches, shall be functionally arranged and clearly identified to facilitate use.

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3.3.4 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall be removable to provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.3.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.3.6 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

3.3.7 Construction. The machine shall be so constructed that when installed and connected to power, it shall be ready for operation. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.8 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. Welding, peening, plugging, or filling with solder or paste shall not be used for reclaiming any defective part.

3.3.9 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.10 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.11 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. External surface edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

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3.3.12 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.13 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3.14 Dials. All feedscrew dials, when furnished, shall be graduated in the inch system. Graduations to indicate stock removal or wheel movement shall be in increments of not more than 0.001 inch. Dial diameters shall be such that the graduations may be easily read. Dials shall be permanently and legibly engraved or etched, and shall have a nonglare finish.

3.3.15 Plates. All words on speed and feed indicating plates and instruction plates shall be in the English language, engraved, embossed, or stamped in plain boldface, easily read letters with contrasting finish.

3.3.16 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

3.3.17 Grinding wheels. Grinding wheels shall conform to MIL-W-17929, be marked in accordance with ANSI B74.13, shall meet the safety requirements designated in ANSI B7.1, and shall be shaped and sized in accordance with ANSI B74.2.

3.4 Detail requirements. Type 0, tool and cutter grinder, bench or pedestal, manual operation, horizontally mounted wheelhead spindle.

3.4.1 Design. Type 0 machines shall consist of a base, with a saddle or slide for wheel column, and a sliding table supporting a swivel table arranged to swivel both sides of center to provide for angle grinding. A swivel wheelhead shall be mounted on a column having vertical travel. The wheelhead shall be provided with means to swivel and lock at any desired angle. The horizontal wheel spindle shall be arranged for mounting straight, cup, or dish type wheels to both ends. Handwheels or quick action levers for operation of transverse and longitudinal movements of the table shall be located so the machine can be operated from operators normal working position. Type 0 machines shall conform to the requirements of table I.

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3.4.2 Base. The base shall have machined ways for either a table or saddle. Machined surfaces shall be included for mounting the column, or transverse ways for the column slide when no saddle is used. The base shall be arranged so it can be mounted either on a bench or a pedestal, or shall be a unit-type construction, as specified (see 6.2.1).

3.4.3 Saddle or column slide. The bottom of the saddle or slide, as applicable, shall be machined and fitted to the base ways on way or roller bearings. The saddle top shall have machined ways for mounting the way rollers or table. The crossfeed of the saddle shall be through a ground-thread crossfeed screw, mounting a dial graduated in increments of not more than 0.001 inch. When specified (see 6.2.1), the saddle shall be equipped with mechanical crossfeed.

3.4.4 Sliding table. The sliding table shall have ways on the bottom, machined and fitted to the bed or saddleways or way rollers. A tee-slot in accordance with ANSI B5.1, or a slot of wedge-type configuration, or table dog rack, shall be provided on the front for carrying the stop-reverse dogs, two of which shall be furnished. The top of the sliding table shall be arranged for mounting the swivel table. All ways and slides shall be protected from the entrance of dust and dirt. The longitudinal table travel shall provide a fast and a slow rate of travel. Drive shall be through a rack and gear. When specified (see 6.2.1), the table shall be equipped with a hydraulic mechanism for longitudinal operation.

3.4.5 Swivel table. The bottom of the swivel table shall be machined to match with and swivel on the top of the sliding table. The top shall be machined and have one or more slots in accordance with ANSI B5.1 extending its entire length for securing tailstocks, headstocks, or other attachments. Graduations shall be provided to indicate angle and taper settings in relationship to the sliding table.

3.4.6 Wheelhead. As specified (see 6.2.1), the wheelhead shall be either single speed or multi-speed. Spindle speeds shall provide approximately 4,400 to 6,000 surface feet per minute wheel speed when wheels 3-1/2 to 6 inches in diameter are used. All pulleys needed to provide these wheel speeds shall be furnished. The wheelhead shall be designed for using straight, cup, or dish wheels of diameters which shall permit full use of the grinding machine's capacity. The wheelhead shall be capable of swiveling a full 360°, and when specified (see 6.2.1), shall be capable of tilting 15° above and below center, both graduated in degrees to indicate the angular setting. The swivel arrangement shall be provided with a clamping device for clamping the wheelhead in any desired position. Raising and lowering of the wheelhead shall be by a

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handwheel and screw. The handwheel shall be graduated in increments of not more than 0.001 inch. The multi-speed spindle shall be a double-end type designed for use of grinding wheels on both ends. All spindles shall be of alloy steel, heat treated and ground. Each spindle shall have a surface finish not rougher than 16 micro-inches with a hardness not less than 46 Rockwell C. Each spindle shall be mounted on pre-loaded ball bearings, sealed and permanently lubricated. The motor shall have protection against the entry of dust, grit, and other injurious substances.

3.4.7 Universal workhead. Unless otherwise specified (see 6.2.1), the machine shall be provided with a manually operated universal workhead capable of swiveling 360° in a horizontal plane, and at least 180° in a vertical plane, both graduated in increments of one degree or less. The workhead spindle shall be of high-grade steel, heat treated and ground, and shall run in anti-friction bearings. Spindle tapers shall be manufacturer's standard. The workhead shall be designed so it can be accurately aligned on the table and securely held in position. The workhead shall be complete with center and draw-in bolt.

3.4.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following standard equipment shall be furnished with each type 0 machine:

(a) Right and left hand tailstocks equipped with centers.

(b) Universal tooth rest, plain tooth rest, blades and all other tooth rests furnished commercially with machine.

(c) Center gage.

(d) Wheel mounts, mandrels, collets, or sleeves normally furnished with machine.

(e) All wrenches, cranks, handles, wheel guards, tee bolts, nuts, and washers normally furnished with the machine.

(f) Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

(g) One diamond of at least .20 carat, complete with holder and bracket.

(h) Reducing collets, the number of each size shall be as specified (see 6.2.1). Tapers shall match universal workhead.

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3.5 Detail requirements. Type I, tool and cutter grinder, floor mounted, manual operation: Style A, horizontally mounted wheel spindle; style B, vertically mounted wheel spindle.

3.5.1 Design. Type I machines shall conform to the requirements of table I. Machines shall be operated from the front and at any position at the side that will facilitate grinding. Handwheels or quick-action levers for operation of transverse and longitudinal movements shall be conveniently located. All moving tables shall travel on ways of construction that will permit easy and free movement. A work table shall be arranged to swivel a full 180° both sides of center, to provide any desired relation between the table and the grinding wheel spindle. The wheelhead spindle shall be designed for mounting straight, cup, and dish grinding wheels. Type I machines shall be either style A or style B, as specified under 1.2. Each style A machine shall consist of a base, sliding table, swivel table, wheelhead, horizontal wheel spindle, driving mechanism, and controls. Each style B machine shall consist of a combination base and pedestal, compound table and swivel arrangement, universal work support fixture, grinding wheel unit, with vertical spindle, driving mechanism, and controls.

3.5.2 Base. The base of each machine shall be rigidly constructed and of sufficient proportions to rest firmly on the floor. It shall be of thick wall construction to preserve alignment. The design shall be such as to offer ample and rigid support for the grinding wheel column and table units. It shall be provided with generously proportioned vee ways, flat ways, or equal means for supporting the saddle or work table arrangement and wheel spindle column. Means shall be provided for preventing exposure of sliding ways and surfaces to dust and grit regardless of the position of the saddle or work table arrangement.

3.5.3 Table workholding arrangement. Each machine shall be designed with one of the following two types of ways as specified (see 6.2.1). One type shall consist of a flat and a vee way accurately fitted to provide smooth and easy travel. The other type shall be the ball track or equal means of anti-friction construction. All ways shall have hardened and ground inserts or be of such construction as will give equally long wearing service. The workholding table shall be provided with a tee-slot in accordance with ANSI B5.1 or comparable means for attaching the workholding device. The workholding table shall be arranged to swivel about a fixed center and shall be provided with graduations for indicating the swivel in degrees. The swivel table for style A machines and the universal work support fixture for style B machines shall be provided with graduations for measuring the swivel in degrees, and at either the left or the right end shall have graduations for measuring

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the swivel in terms of taper in inches per foot. Means for clamping the swivel table in the desired location shall be provided. Each style A machine shall include in its design a sliding table arrangement in combination with a swivel table, with provisions for moving the workpiece past the grinding wheel to facilitate grinding of cutters. Each style B machine shall accomplish the identical function by moving the grinding wheel past the workpiece.

3.5.4 Wheelhead. The grinding wheelhead shall be of the self-contained type. The wheel spindle shall be of high grade steel, ground and mounted on precision, preloaded, antifriction bearings with sealed-in lifetime lubrication or cartridge-type mounting. The spindle shall be of a type to permit the use of grinding wheels. The wheelhead shall swivel in a horizontal plane for style A machines and in a vertical plane for style B machines. When specified (see 6.2.1), the wheelhead shall be capable of tilting 15° above and below center, both graduated in degrees to indicate the angular setting. The wheelhead shall give vibrationless operation up to the maximum RPM of the grinder. Unless otherwise specified (see 6.2.1), two or more changes of wheel speed shall provide 4,400 to 6,000 surface feet per minute on wheels 3-1/2" to 6" in diameter. The construction of the wheelhead shall be such as to permit the removal of the entire spindle assembly from the machine as a unit for convenience in servicing. The grinding wheelhead shall be adjusted vertically in a housing fixed to the bed or wheel slide. The housing or other means provided for support of the wheelhead shall be equipped with a flexible bellows or equal means to prevent entry of grit and dust.

3.5.5 Wheel spindle drive. The wheel spindle shall be driven by a motor. The motor arrangement shall have means for maintaining proper belt tension, if a belt other than a timing type belt is used.

3.5.6 Mechanical controls. The machine shall have all controls convenient to the operator, accessible from any working position. The controls shall be arranged for operating the grinder at the front, right hand side or left hand side of the machine. Style A machines shall be equipped with a traveling table for longitudinal movement to provide means for sharpening the cutters involved. Style B machines shall be equipped with a traveling grinding wheel spindle unit to accomplish the same function. The transverse movement of the workholding unit or saddle shall be by handwheel conveniently located and provided with graduations reading to 0.001" or less.

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3.5.7 Transverse movement. The transverse movement shall be provided either by a wheel-spindle column supported on a base which moves the wheel-spindle unit into the work, or by a saddle unit which moves the work-supporting table into a fixed wheelhead unit. Either unit shall be mounted directly on the base ways and shall have ample bearing surface to assure permanently accurate alignment and shall be provided with adjustable gibs or equal means to compensate for wear as required because of manufacturer's design. The transverse movement to either saddle or wheelhead shall be operated by a handwheel having a graduated dial to read in increments of 0.001 inch or less.

3.5.8 Universal workhead. Unless otherwise specified (see 6.2.1), type I, style A machines shall be provided with a manually operated universal workhead capable of swiveling 360° in a horizontal plane and at least 180° in a vertical plane, both graduated in increments of one degree or less. The workhead spindle shall be of high-grade steel, hardened and ground and shall run in antifriction bearings. Spindle tapers shall be manufacturer's standard. The workhead shall be designed so it can be accurately aligned on the table and securely held in position. The workhead shall be complete with center and draw-in bolt.

3.5.9 Standard equipment. Unless otherwise specified (see 6.2.1), the following standard equipment shall be furnished with each type I machine:

(a) Right and left hand tailstocks equipped with centers. Style B machines shall have an attachment equipped with centers, fitted to the universal work support fixture.

(b) Universal tooth rest, plain tooth rest, blades, and all other tooth rests furnished commercially with machine.

(c) Center gage.

(d) Wheel mounts, mandrels, collets, or sleeves, normally furnished with machine.

(e) All wrenches, cranks, handles, wheel guards, tee bolts, nuts, and washers, normally furnished with the machine.

(f) Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

(g) One diamond of at least .20 carat, complete with holder and bracket.

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(h) Reducing collets, the number of each and sizes shall be as specified (see 6.2.1). Tapers shall match universal workhead for style A machines or the universal work support fixture for style B machines.

3.6 Detail requirements. Type II, tool and cutter grinder, floor mounted, power operated, horizontal wheelhead spindle.

3.6.1 Design. The type II grinding machine covered by this specification shall be suitable for all varieties of cutter grinding within its capacity. The cutter grinding capability shall include milling cutters, counterbores, reamers, and other metal-cutting tools. Each machine shall be equipped with a power operated, revolving workhead (or headstock), and attachments for cylindrical, surface, internal grinding, and gear cutter sharpening. In addition to fast and slow hand longitudinal feed, power longitudinal feed, integral with the machine, shall be provided. Type II machines shall conform to the requirements of table I.

3.6.2 Base. The base shall be of cabinet-type construction with ribs to insure retention of accuracy within the limits specified herein. The base shall have one vee way and one flat way for mounting the sliding table, or antifriction ways. Means shall be provided to cover the ways, regardless of the position of the table, while in normal operation.

3.6.3 Sliding table. The sliding table shall have one flat and one vee way to match the table ways on the base or to fit the way rollers when antifriction ways are used. Reversing of the table shall be by means of dogs adjustable along a tee-slot or table dog rack on the front of the table and provided with adjustment screws. At least two dogs shall be provided. The top of the table shall be machined for mounting the swivel table.

3.6.4 Swivel table. The swivel table shall be mounted on the sliding table and be arranged for angular adjustment up to 90° in both directions. A pivot pin or other means shall be used to maintain a constant point of pivot. Means for clamping shall be provided. Graduations shall be provided to indicate settings in degrees and inches per foot. A fine screw adjustment shall be provided for taper settings up to not less than 2 inches per foot, indicated by a scale at the end of the table. At least one slot in accordance with ANSI B5.1 shall be provided in the top surface of the table for fixture location and bolting.

3.6.5 Wheelhead. The wheelhead unit shall be slide mounted on the vertical ways or round column of the wheelhead column (upright). Vertical adjustment shall be through an elevating handwheel. The handwheel shall be graduated in increments of 0.001 inch or less. The wheel spindle shall be arranged for mounting straight or cup wheels on both ends.

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The spindle assembly shall be of the cartridge type using preloaded ball bearings. All diameters of the spindle, except threaded diameters, shall be ground to insure spindle balance. The wheelhead column shall swivel at least 110° both sides of zero and when specified (see 6.2.1), the wheelhead shall be capable of tilting 15° above and below center, both with settings indicated by graduations in degrees. A wheel spindle motor shall be mounted in or on the wheelhead and shall be provided with means for maintaining belt tension, if a timing type belt is not used.

3.6.6 Transverse movement. The transverse movement shall be provided either by a wheel spindle column supported on a base which moves the wheel spindle unit into the work, or by a saddle unit which moves the work-supporting table into a fixed wheelhead unit. Either unit shall be mounted directly on the base and shall have ample bearing surface to assure permanently accurate alignment and shall be provided with adjustable gibs or equal means to compensate for wear where required because of manufacturer's design. The transverse movement to either saddle or wheelhead shall be operated by a handwheel provided with graduated dial to read in increments of 0.001 inch or less.

3.6.7 Headstock. The headstock shall be a motor driven, independent unit, and shall be adjustable along the table and removable therefrom. It shall have a horizontal swivel adjustment of not less than 90° both sides of zero alignment with the table, indicated by graduations in degrees. The headstock spindle shall be of alloy steel and shall be ground and have a surface hardness of not less than Rockwell 48 on the C scale. It shall be mounted in preloaded ball or roller bearings, arranged for either dead or live center grinding. The taper shall be the manufacturer's standard, fitted with a center. The spindle shall be arranged for mounting faceplates and chucks. The headstock shall have four or more speeds within the range of 128 RPM or less to 400 RPM or more. The headstock shall be furnished with a live center pulley.

3.6.8 Tailstock. The tailstock shall be designed for positioning along the table in alignment with the headstock when the headstock is set on zero. The tailstock taper, or other internal configuration, shall be as specified (see 6.2.1). The spindle shall act against a spring to allow tension adjustment. A clamping device shall be provided for locking the spindle with desired tension.

3.6.9 Feeds and speeds. The speeds of the wheel and workhead shall be independent of each other. Not less than three changes of wheel speeds shall be provided to produce speeds of 4,400 to 6,000 surface feet per minute for wheels 3-1/2 to 6 inches in diameter. The cross-feed mechanism shall be hand operated with a fine screw adjustment having

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a dial graduated in increments of 0.001 inch stock removal or less. The method of table feed may be either mechanical or hydraulic and shall have sufficient travel to grind work equal in length to the specified table travel in accordance with table I.

3.6.10 Controls. The grinder shall be provided with all controls in convenient reach of the operator from the front of the grinder. Controls shall be provided for selecting rates of power travel on the table. Means shall be provided for selecting fine and coarse crossfeed. The table reversing lever shall be operable either manually or by the table reversing dogs.

3.6.11 Universal workhead. Unless otherwise specified (see 6.2.1), the machine shall be provided with a manually operated universal workhead capable of swiveling 360° in a horizontal plane, and at least 180° in a vertical plane, both graduated in increments of one degree or less. The workhead spindle shall be of high-grade steel hardened and ground and shall run in antifriction bearings. Spindle tapers shall be manufacturer's standard. The workhead shall be designed so it can be accurately aligned on the table and securely held in position. The workhead shall be complete with center and draw-in bolt.

3.6.12 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each type II machine:

(a) Three grinding centers.

(b) Set of work driving dogs.

(c) Left and right hand tailstocks.

(d) All tooth rests, tooth rest bars, holders, tooth rest holder clamps, blades, and other tooth rest equipment normally furnished with the grinder.

(e) Wheel sleeves, mounts, collets, or mandrels normally furnished with machine.

(f) Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

(g) Center height gage.

(h) Reducing collets, the number of each and sizes shall be as specified (see 6.2.1). Tapers shall match universal workhead and headstock.

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(i) All wrenches, cranks, handles, clamps, cutter bars, wheel guards, wheel pullers, tee bolts, nuts, and washers normally furnished with the grinder.

(j) One diamond of at least .20 carat, complete with holder and bracket.

(k) One cylindrical grinding attachment, complete with 3-jaw universal chuck, 2 sets nonreversible chuck jaws (one internal and one external) chuck adapter, spindle assembly, driving motor and all other associated components required. Chuck size, spindle taper, and motor horsepower shall be as specified (see 6.2.1).

(l) One internal grinding attachment complete with mounting bracket, wheel spindle housing and spindle (for use with removable quills) driven by wheelhead spindle through a pulley and belt. The number of quills and their sizes, and the number of grinding wheels, their sizes, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

(m) One surface grinding attachment, complete with swivel vise (360° in both horizontal and vertical planes), intermediate support, base plate, and necessary tee bolts.

(n) One gear cutter sharpening attachment including bracket, gage, pawl, necessary tee bolts, and cutter bushings. The number and size cutter bushings shall be as specified (see 6.2.1).

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TABLE I
SIZES AND CAPACITIES

CHARACTERISTIC	REQUIREMENT[1]								
	TYPE O		TYPE I				TYPE II		
			STYLE A			STYLE B			
	Size 615	Size 817	Size 815	Size 1022	Size 1227	Size 815	Size 815	Size 1022	Size 1227
CAPACITY									
Swing over table, inches	6 1/4	8	8	10	12	8	8	10	12
Distance between centers, inches									
of tailstocks	15	17	15	22	27	15	15	22	27
of universal workhead and tailstock	10	12	11	20	24	-	11	28	24
of headstock and tailstock	-	-	-	-	-	-	11	22	24
SLIDING TABLE & SWIVEL TABLE ARRANGEMENT									
Table working surface, inches									
length by width	22 1/2 x 2 7/8	24 x 4	24 x 4	30 x 5	38 x 5	16 x 16	24 x 4	30 x 5	38 x 5 1/2
Table longitudinal movement, inches	10	10	10	16	16	14	10	16	16
Table longitudinal movement actuated by,	Hand	Hand	Hand	Hand	Hand	Hand	-----	Hand and Power	-----
Table power longitudinal movement, mechanical or hydraulic									
Feeds number of	-	-	-	-	-	-	Infinite	Infinite	Infinite
Feed rates, inches per minute	-	-	-	-	-	-	7 to 90	7 to 90	2 to 200
Table cross movement, inches	4 3/4	6 1/2	4 1/2	8	8	14	8	8	8
Swivel table horizontal swivel, degrees each side of zero	90	90	90	90	90	180	90	90	90
COMPOUND TABLE & SWIVEL ARRANGEMENT									
Carriage cross travel, inches	-	-	-	-	-	14	-	-	-
Carriage angular adjustment, degrees	-	-	-	-	-	180	-	-	-
Carriage longitudinal adjustment, inches	-	-	-	-	-	10	-	-	-
Table angular adjustment in vertical plane, degrees	-	-	-	-	-	45	-	-	-
Base table vertical adjustment, inches	-	-	-	-	-	12 1/2	-	-	-
Center table size, inches, length x width	-	-	-	-	-	24 x 3 1/2	-	-	-
WHEELHEAD OR GRINDING UNIT									
Spindle speeds, number	Single or	Multiple[2]	2	2	4	3	3	3	4
Wheelhead vertical adjustment, inches	6 1/2	6 1/2	7	8	12	-	7	8	12
Grinding wheel vertical adjustment, inches	-	-	-	-	-	1 1/2	-	-	-
Wheelhead horizontal swivel, degrees each side of zero	110	110	110	110	110	-	110	106	110
Angular adjustment of spindle head, degrees	-	-	-	-	-	90	-	-	-
Horizontal travel of spindle head, inches	-	-	-	-	-	7	-	-	-
Vertical tilt, degrees above and below center	15	15	-	15	15	-	15	15	15
Motor horsepower	1/2	1/2	1/2	3/4	1	1	3/4	3/4	1
POWER OPERATED HEADSTOCK OR WORKHEAD									
Speeds, number of	-	-	-	-	-	-	4	4	4
Speed range, RPM	-	-	-	-	-	-	128 to 400	128 to 400	128 to 400
Horizontal swivel, degrees each side of zero	-	-	-	-	-	-	90	90	90
Motor horsepower	-	-	-	-	-	-	1'6	1'4	1/4

1/ Unless otherwise specified (see 6.2.1).
Requirements are not less than those shown. Where a range is shown, the performance shall encompass a range from the stated minimum or less to the stated maximum or more.

2/ As specified under 3.4.6.

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3.7 Detail requirements. Type III, tool, cutter, gashing, and radius grinder, floor mounted, vertically and horizontally mounted wheelhead spindle, clockwise from vertical 105°, counterclockwise from vertical 105°.

3.7.1 Design. The tool, cutter, gashing, and radius grinders covered by this specification shall be suitable for resharpening conventional type of cutters, including end mills, reamers, counterbores, form cutters, etc., with straight, tapered, or helical teeth, and involving convex or concave radii. Construction shall be such that all operations necessary in sharpening of most types of cutters can be performed in one setting or chucking of the workpiece, thus assuring concentricity of the cutting edges with the shank of the cutter. The machine shall be a compact self-contained unit, and shall consist essentially of a box-type bed on the top of which shall be mounted the wheelhead unit, and the workhead unit. The wheelhead unit shall provide support for the grinding wheel spindle, and be equipped with the necessary gib adjusted slides to provide for adjustment of the grinding-wheel spindle in the transverse, and vertical directions. All slides shall be fitted with handwheels and the transverse and vertical slides shall have dials graduated to not more than 0.001 inch movement to effect precision positioning of the grinding wheel. Adjustable stops shall be provided on transverse slides to effect precision limits of travel in this direction. The workhead unit shall be supported on the base of the machine in precision antifriction bearings so that the whole workhead unit can be swiveled in a horizontal plane through at least 210°, fitted with adjustable stops to limit the swivel in both directions. On top of this workhead swivel member shall be precision fitted compound slides to provide adjustment of the workholder in transverse and longitudinal directions. The swivel member shall be mounted on a slide to provide for adjustment at the longitudinal axis. Positional adjustment of the offset slides shall be by means of an accurately calibrated scale, graduated in increments of 0.001 inch or less of movement, and a dial handwheel or crank, or equal means. Type III machines shall conform to the requirements of table II; option shall be furnished only when specified.

3.7.2 Base. The base shall be of cabinet-type construction which shall provide support for the wheelhead unit and the workhead unit. It shall be provided with a built-in compartment for storage of tools, grinding wheels, and accessories.

3.7.3 Wheelhead unit. The wheelhead unit shall consist of sliding members, permitting transverse and vertical adjustment, and shall be capable of swiveling 105° clockwise and counterclockwise from its vertical position. The lowest member shall provide the cross movement.

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Above this member shall be provided a vertical column and slide which can be raised or lowered by a handwheel and screw arrangement or equal means. The grinding wheel shall be mounted on an individual spindle, which shall be supported on precision antifriction bearings mounted on a swiveling member. Slides shall be provided with necessary gibs for maintaining adjustment. The slides shall be provided with means for lubrication and shall be protected against the entry of grinding dirt and dust. The wheelhead unit shall be fitted with a permanently mounted clearance angle scale to provide quick setting of the grinding wheel for any desired clearance angle in both the vertical and transverse directions. The grinding-wheel motor shall be mounted on the vertical slide. Adjustable wheel guard shall be furnished for the grinding wheel. The spindle shall provide vibrationless operation up to the maximum RPM of the grinder. Spindle speeds shall provide 2,800 to 5,600 surface feet per minute wheel speed when using wheels 3 inches to 6 inches in diameter.

3.7.4 Work head unit. The work head unit shall be of the type to revolve about the axis at the cutter an amount equal to the helix angle of the cutter and the machine configuration so that a gash can be made to center with negative, positive, and radial rake, and adjusted to secure the desired rake. The work head unit or cutter-holding unit shall be mounted on a swivel base which shall permit the swiveling of the work head through at least 210° in a horizontal plane. Adjustable stops shall be provided to limit the swivel in both directions. The swiveling base shall carry a screw-operated cross-slide. On the cross-slide shall be mounted a second slide, known as the top slide, which will carry the work head or cutter head. The top slide shall also be screw-operated, and shall be provided with means for adjustment of the work head toward and away from the center of the swivel. The top slide shall have tee-slots permitting work head to be set in any longitudinal position to suit the length of work. A positive stop shall be provided to bear against the rear surface when the head is centrally located. The work holder shall have means for indexing 2, 3, 4, 6, and 8 flute mills. The work holder shall be capable of pivoting the axis at the cutter $\pm 20^{\circ}$ from horizontal position. A new index plate shall be required for mills other than described above. Graduations shall be provided for setting to clearances desired when grinding end mills. A work holding spindle shall be mounted on antifriction bearings and shall accept adapters for reducing collets as specified (see 6.2.1). Tapers shall be manufacturer's standard and shall conform to ANSI B5.10. Unless otherwise specified (see 6.2.1), a tracer grinding and turning attachment shall be furnished.

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3.7.5 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each type III machine. All equipment shall be adaptable and suitable for the machine furnished.

(a) All necessary wrenches, cranks, handles, wheel guards, tee bolts, nuts, and stock supports normally furnished as standard equipment with the machine.

(b) Borazon grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

(c) Ball bearing work holding spindle bored to accept adapters:

(1) for #5 B&S taper shank.

(2) for #7 B&S taper shank.

(3) for #9 B&S taper shank.

(4) for straight shank reducing collets:

a. for sizes 1/8" to 3/8".

b. for sizes 3/8" to 1".

c. for sizes 3/4" to 2-5/32".

(d) Two 3-lobe cams, one for gashing, one for relieving.

(e) Set of index plates for 2, 3, 4, 6, and 8 flute cutters.

(f) Set of 20 reducing collets - 1/8 inch, 3/16 inch, 1/4 inch, 5/16 inch, 3/8 inch, 7/16 inch, 1/2 inch, 5/8 inch, 3/4 inch, 7/8 inch, 1 inch, 1-1/8 inches, 1-1/4 inches, 1-3/8 inches, 1-1/2 inches, 1-5/8 inches, 1-3/4 inches, 1-7/8 inches, 2 inches, and 2-5/32 inches.

(g) Preset fixture.

(h) #5 B&S taper shank adapter.

(i) #7 B&S taper shank adapter.

(j) #9 B&S taper shank adapter.

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(k) Adapter for reducing collets from 1/8" to 3/8".

(l) Adapter for reducing collets from 3/8" to 1".

(m) Adapter for reducing collets from 3/4" to 2-5/32".

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TABLE II.

Size and Capacity

Characteristic	Requirement Type III Size 610
Swing over main slide, inches	6
Maximum mill diameter (for head with #9 B&S taper spindle) over table, inches	8
Tool length accommodated (length of flutes), inches	10
Collet work diameter, range, inches	1/8 to 2-5/32
Longitudinal movement of main (work) slide, inches	5-3/4
Wheelhead, slide travel, inches	
Vertical	6
Cross	6
Longitudinal	0
Angular range of workhead:	
In horizontal plane (workhead swivel base), degrees	278
In vertical plane (workhead angular adjustment) cutter axis angular range, degrees	±20
Workhead spindle speed, RPM	Manual or 440 1/
Wheelhead spindle motor, horsepower	1
Optional horsepower	2
Workhead spindle operation	Manual
Tracer grinding and turning attachment, horsepower	1/3

1/When tracer grinding and turning attachment is used.

Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

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3.8 Detail requirements. Type IV, tool, cutter, flute and radius grinder, floor mounted, horizontally mounted wheelhead spindle.

3.8.1 Design. The tool, cutter, flute and radius grinders covered by this specification shall be suitable for resharpening conventional types of cutters, including end mills, reamers, counterbores, form cutters, etc., with straight, tapered, or helical teeth, and involving convex or concave radii; and for making new cutters. Construction shall be such that all operations necessary in the production and sharpening of most types of cutters can be performed in one setting or chucking of the workpiece, thus assuring concentricity of the cutting edges with the shank of the cutter. The machine shall be a compact self-contained unit, and shall consist essentially of a box-type bed on the top of which shall be mounted the wheelhead unit, and the workhead unit. The wheelhead unit shall provide support for the grinding wheel spindle, and be equipped with the necessary gib adjusted slides to provide for adjustment of the grinding-wheel spindle in the transverse, longitudinal, and vertical directions. All slides shall be fitted with handwheels, and the transverse and vertical slides shall have dials graduated to show not more than 0.001 inch increment movement to effect precision positioning of the grinding wheel. Adjustable stops shall be provided on transverse and longitudinal slides to effect precision limits of travel in those directions. The workhead unit shall be supported on the base of the machine in precision antifriction bearings so that the whole workhead unit can be swiveled in a horizontal plane through at least 210°, fitted with adjustable stops to limit the swivel in both directions. On top of this workhead swivel member shall be a precision fitted gib adjustable offset slide to provide displacement of the axis of the workhead spindle to the front and rear of the horizontal swivel point. Positional adjustment of the offset slide shall be by means of an accurately calibrated scale, graduated to show movement increments of 0.001 inch or less, and a dial handwheel or crank, or equal means. Type IV machines shall conform to the requirements of table III.

3.8.2 Base. The base shall be of cabinet-type construction which shall provide support for the wheelhead unit and the workhead unit. It shall be provided with a built-in compartment for storage of tools, grinding wheels, and accessories.

3.8.3 Wheelhead unit. The wheelhead unit shall consist of three sliding members, permitting longitudinal, transverse, and vertical adjustment. The lowest member shall provide the cross movement. Above this member shall be provided a vertical column and slide which can be raised or lowered by a handwheel and screw arrangement or equal means. The grinding

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wheel shall be mounted on an individual spindle, which shall be supported on precision antifriction bearings mounted on a longitudinal slide operated by a handwheel or handcrank, through a rack and pinion or equal means. Slides shall be provided with necessary gibs for maintaining adjustment. The slides shall be provided with means for lubrication and shall be protected against the entry of grinding dirt and dust. The wheelhead unit shall be fitted with a permanently mounted clearance angle scale to provide quick setting of the grinding wheel for any desired clearance angle in both the vertical and transverse directions. The grinding wheel motor shall be mounted on the vertical slide and be belt connected to the grinding wheel spindle. The belt shall be guarded by a suitable cover, and an adjustable wheel guard shall be furnished for the grinding wheel. The spindle shall give vibrationless operation up to the maximum RPM of the grinder. Spindle speeds shall provide 4,000 to 5,400 surface feet per minute wheel speed when using wheels 3 inches to 4 inches in diameter.

3.8.4 Workhead unit. The workhead unit on Size 84 machines shall be of the plain type. Size 126 machines shall have a spiral grinding arrangement.

3.8.4.1 Workhead unit (plain type for size 84). The workhead unit or cutter holding unit shall be mounted on a swivel base which shall permit the swiveling of the workhead through at least 210° in a horizontal plane. Adjustable stops shall be provided to limit the swivel in both directions. The swiveling base shall carry a screw operated cross-slide. On the cross-slide shall be mounted a second slide, known as the top slide, which will carry the workhead, or cutter head. The top slide shall also be screw operated, and shall be provided with means for adjustment of the workhead toward and away from the center of the swivel. The top slide shall have tee-slots permitting workhead to be set in any of three longitudinal positions to suit the length of work. A positive stop shall be provided to bear against the rear surface when the head is centrally located. The work holder shall have means for indexing 2, 3, and 4 flute mills. A tooth rest shall be provided for mills with more than four flutes. Graduations shall be provided for setting to clearances desired when grinding end mills. A work holding spindle shall be mounted on antifriction bearings and shall be bored for a taper shank; it shall be provided with a knockout rod, and reducing collets as specified by the procuring agency (see 6.2.1). Tapers shall be manufacturer's standard and shall conform to ANSI B5.10. Unless otherwise specified (see 6.2.1), a tracer grinding and turning attachment shall be furnished.

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3.8.4.2 Workhead unit (with helix grinding arrangement for size 126). The workhead unit shall be supported on the base of the machine in precision antifriction bearings so that the whole workhead unit can be swiveled in a horizontal plane. Adjustable stops shall be provided to limit the swivel in both directions. On top of the workhead horizontal swivel member shall be a precision fitted gib adjustable offset slide to provide displacement of the axis of the workhead spindle to the front and rear of the horizontal swivel pivot point. Positional adjustment of the offset slide shall be by means of an accurately calibrated scale and a dial handwheel or equal, graduated in increments of 0.001 inch or less. Above the offset slide shall be located the main slide of the workhead unit and it shall be precision fitted and gib adjusted. Movement of the main slide shall be effected by either of two handwheels, one located on the side of the main slide. The main slide shall support the workhead housing mounted so as to be adjustable in a vertical plane, above and below a horizontal position. The workhead spindle shall be mounted on the main slide with a suitable mechanism operating through a spiral lead device. A set of index plates for 8, 10, and 12 equally notched spaces shall be provided. A workhead spindle shall also be fitted with a permanently mounted, direct indexing disk, spring loaded micrometer, and an adjustable index finger for ratchet indexing of 10, 12, and 14 evenly spaced notches. The workhead spindle shall be mounted on precision antifriction bearings and shall be made to accommodate split collets. An adjustable toothrest shall be provided for mounting on the grinding wheel spindle housing. The workhead unit shall be equipped with a permanently mounted dust protected motor for effecting rotation of the workhead spindle by belt drive. A set of three index plates for helical lead mechanism index, 8, 10, and 12 divisions, shall be furnished with the workhead unit. All slides shall be provided with a one shot lubricating system.

3.8.5 Standard equipment, type IV, size 84 (plain type workhead unit). Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine. All equipment shall be adaptable and suitable for the machine furnished.

(a) All necessary wrenches, cranks, handles, drive pulleys, drive belts, wheel guards, tee bolts, nuts, and stock supports normally furnished as standard equipment with the machine.

(b) Cutter tooth setting gage.

(c) Tooth rest and holder.

(d) Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

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(e) One diamond of at least .20 carat, complete with holder and bracket.

(f) Ball bearing work holding spindle bored one of the following ways, as specified (see 6.2.1):

(1) For #9 B&S taper shank, with 9 to 7 and 9 to 5 reducing collets.

(2) As required by the procuring agency.

3.8.6 Standard equipment, type IV, size 126 (helix grinding workhead). Unless otherwise specified (see 6.2.1), the following standard equipment shall be furnished with each machine. All equipment shall be adaptable and suitable for the machine furnished.

(a) All necessary wrenches, cranks, handles, drive pulleys, drive belts, wheel guards, tee bolts, nuts, and stock supports normally furnished as standard equipment with the machine.

(b) Tooth rest.

(c) Tooth rest centering gages.

(d) Set of three index plates for helix lead mechanism indexing device, 8, 10, and 12 divisions.

(e) Set of 13 straight cylindrical collets with adapters as required - 1/8 inch, 3/16 inch, 1/4 inch, 5/16 inch, 3/8 inch, 7/16 inch, 1/2 inch, 5/8 inch, 3/4 inch, 7/8 inch, 1 inch, 1-1/8 inches, and 1-1/4 inches.

(f) Set of six taper collets - the O.D. of collets shall fit the workhead spindle. The size of work holding taper shall be as specified (see 6.2.1).

(g) Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

(h) One diamond of at least .20 carat, complete with holder and bracket.

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TABLE III.
Sizes and capacities

Characteristic	Requirement 1/	
	Type IV	
	Size 84	Size 126
Swing over main slide, inches	8-5/8	12
Maximum tool length accommodated, inches	4-1/2	6
Collet work diameter, range, inches	1/4 to 1-1/4	1/8 to 1-1/4
Longitudinal movement of main (work) slide, inches	3-1/4	5-3/4
Wheelhead, slide travel, inches		
Vertical	5-3/8	8-3/4
Cross	6-1/4	7-1/2
Longitudinal	4-1/2	6
Angular range of workhead degrees:		
In horizontal plane (workhead swivel base)	210	235
In vertical plane (workhead angular adjustment)	30	40
Workhead spindle speed, RPM	manual or 440 2/	285
Wheelhead spindle motor, horsepower	1/3	1/2
Workhead spindle motor, horsepower	manual	1/15
Tracer grinding and turning attachment, horsepower	1/3	

1/ Unless otherwise specified (see 6.2.1). Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

2/ When tracer grinding and turning attachment is used.

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3.9 Performance. Each machine shall be capable of resharpening conventional type cutters to like new condition tolerances and finishes. Radial runout shall be not more than 0.0015 inch TIR. Cutter tooth face edges shall be parallel within 0.0015 inch with not rougher than 16 microinch finish.

3.10 Alignment tolerances. Each machine shall meet the alignment tolerances of tables VI or VII as applicable.

3.11 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical system shall conform to Machine Tool Electrical Standards of the National Electrical Manufacturers' Association. Undervoltage protection shall be provided.

3.11.1 Motors. Motors shall have ball or roller bearings. Horsepower ratings shall conform to the requirements shown in tables I, II, and III. Unless otherwise specified (see 6.2.1), motors shall be continuous duty, squirrel cage induction type, 230/460 volts, 3 phase, 60 Hertz, AC, initially wired for 230 volt operation. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements of a dripproof enclosure.

3.11.2 Motor nameplate. Each motor shall be equipped with the motor manufacturer's nameplate bearing the motor manufacturer's name, the motor model number, the serial number if any, rated horsepower, speed in revolutions per minute, and the motor electrical characteristics. With the motor mounted on the machine, the nameplate shall be easily visible without removing or turning the motor.

3.12 Coolant system. When specified (see 6.2.1), a flood type or mist type coolant system shall be provided. The system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit full flow of coolant, and shall have means for draining and cleaning. The system shall include a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant flow.

3.13 Hydraulic system. When hydraulics are included in the design of the machine, a hydraulic system shall be provided. The system shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

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3.14 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1). Optional equipment furnished shall be adaptable and suitable to the type and size grinding machine furnished.

3.15 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for class IIB equipment.

3.16 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.17 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application.

Service interval.

Type of lubricant.

Viscosity.

Military or Federal Specification (if available).

3.18 Identification marking.

3.18.1 Civil agencies. Each machine shall be marked in a plain and permanent manner with the model and serial number and with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.18.2 Military agencies. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

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Nomenclature.

Manufacturer's name.

Manufacturer's model designation.

Manufacturer's serial number.

Type _____ size _____ swing _____ inches by _____ inches center to center.

Power input (volts, total amps, phases, cycles).

Contract Number or Order Number.

Federal Stock Number or Plant Equipment Code.

Date of manufacture.

U S

3.19 Identification plate. When specified (see 6.2.1), a separate identification plate of corrosion-resistant metal shall be securely attached to the machine. The plate shall be permanently and legibly captioned "Departmental or Activity Identification" in letters not less than 3/16 inch high and shall have blank space for stamping 25 characters of like size.

3.20 Technical data.

3.20.1 Civil procurements. For procurements by Civil agencies, two copies each of the manufacturer's handbook of instructions for machine operation, care, and maintenance, a machine parts list, and two copies each of foundation and anchor bolt plans and wiring diagrams shall be furnished.

3.20.2 Military procurements. Data shall be furnished as specified (see 6.2.2).

3.21 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.2, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise examination for compliance with the applicable requirements in 3.3 through 3.21, examination to ensure compliance with the additional safety and health requirements specified by the procuring activity under 6.2.1(e), and all tests in 4.4. Failure to pass any examination or any test shall be cause for disapproval of the First Article.

4.3 Inspection for acceptance. Each machine shall be examined in accordance with 4.3.2 and tested in accordance with 4.4 through 4.4.3. Unless otherwise specified (see 6.2.1), additional inspection shall be made in accordance with the provisions set forth in MIL-STD-105, as specified in 4.3.3, 4.3.4, and 4.3.5. The inspection terms and definitions shall be interpreted in accordance with MIL-STD-109. Results shall be kept available for review by Government inspectors.

4.3.1 Inspection of materials and components. The supplier shall ascertain that components and materials used were manufactured, tested, and inspected to insure that machines can be produced in accordance herewith.

4.3.2 Examination for safety and health requirements. Each machine shall be examined to ensure compliance with 3.3.4 and with the additional safety and health requirements specified by the procuring activity under 6.2.1(e) (see 6.5). Each machine shall comply with all such requirements.

4.3.3 Examination of size and capacity. Examination shall be made of each end item in the sample to determine compliance with the requirements of tables I, II, or III, as applicable. Any characteristic which does not comply with the applicable table shall be classified as a defect. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

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TABLE IV.
Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Type of finish not as specified or blistered, peeled, chipped, or thin film.		X
	Not finished where required.	X	
	Component or assembly that is painted or otherwise coated where serviceability may be affected, e.g., paint or gasket.	X	
Material	Component not fabricated of the specified material.	X	
Design	Any characteristic not in accordance with the specified requirements.	X	
Construction and workmanship, general (applicable to all components and assemblies)	Part missing or not specified type or size.	X	
	Fractured, split, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired affecting serviceability.	X	
	Defective or otherwise damaged but not affecting serviceability.		X
	Misplaced, loose, or not in proper alignment affecting serviceability.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel or may affect serviceability.	X	
	Operation omitted or not properly performed, affecting serviceability.	X	
	Operation omitted or not properly performed, but not affecting serviceability.		X
	Functioning component that requires abnormal force to operate.		X

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TABLE IV.
Classification of defects (cont.)

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship (cont.)	Not connected or joined as specified or poorly accomplished, affecting serviceability.	X	
	Any adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended.	X	
	Any component not readily accessible for servicing where required.		X
Welding and brazing (where required)	Not type specified, missing, incomplete, burn holes, cracked, fractured, otherwise not fused.	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed.		X
Soldering (where applicable)	Missing, not adherent or incomplete.	X	
	Not clean (flux or flux residue not removed) not smooth (surface not finished neatly) or pin holes in solder.		X
Bolts, nuts, screws-studs, and other types of threaded fasteners	Missing, broken, stripped, fractured, or loose.	X	
Electrical assembly	Wiring cut, adraded, not properly joined, loose at terminal or not inclosed in specified conduit where required.	X	
	Adequate slack not provided for wiring to relieve strain, or excessive insulation stripped from wiring.		X

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TABLE IV.
Classification of defects (cont.)

Examine	Defect	Classification	
		Major	Minor
Lubrication (where applicable)	Not lubricated where required.	X	
	Insufficiently lubricated.		X
	Not specified type lubrication fitting, affecting serviceability.	X	
	Lubrication fitting not readily accessible for servicing.		X
Assembly	Component not properly assembled or secured but not affecting serviceability, e.g., lockwasher missing from metal fastener.		X
	Unit preceptibly out of square or alignment.	X	
Marking identification, instructions, manual	Missing, incomplete, not legible.		X

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4.3.4 End product visual examination. Each completely assembled end item in the sample shall be examined for defects in finish, material, construction, workmanship, and marking. Defects shall be classified as outlined in table IV for recording of inspection results and corrective action. Inspection level shall be S-2 with an AQL of 4.0 defects per hundred units for major defects, and 6.5 defects per hundred units for minor defects.

4.3.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table V. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery and selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE V.
Classification of Preparation for Delivery Defects

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of container

4.4 Tests. Unless otherwise specified (see 6.2.1), each grinding machine shall be tested as specified in 4.4.1 through 4.4.3. Failure of any machine to pass any test shall be cause for rejection.

4.4.1 Operational test. Each machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be proved during the trial period.

4.4.2 Performance test. Each grinding machine shall be tested as follows:

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4.4.2.1 Type 0, I, and II machines. Grind or regrind a 4 by 1-1/2 inch, high speed steel, left or right hand, arbor mounted type, helical, plain milling cutter having a minimum of ten cutting teeth. Finished cutter physical dimensions shall conform to ANSI B5.3 for standard milling cutters. Tolerances shall be in accordance with 4.4.2.3.

4.4.2.2 Type III and IV machines. Grind or regrind a 1 by 1-5/8 inch, multiple fluted, high speed steel, left or right hand, shank mounted type, helical end mill. Shank tapers shall conform to ANSI B5.10. Finished cutter physical dimensions shall conform to ANSI B5.3. Tolerances shall be in accordance with 4.4.2.3.

4.4.2.3 Tolerances. Diameter of cutter tooth edges, when measured at any point along their full length, shall be within +0.000 -0.0015. Runout shall not exceed 0.0015 TIR. Surface finish shall be not rougher than 16 microinches.

4.4.3 Alignment tests. Each machine and its components shall be tested for conformance with the accuracies specified in tables VI or VII, as applicable. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

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TABLE VI.
Alignment tests (types 0, I, and II)

Test to be applied, as applicable, to type of machine	Maximum, allowable deviation
Worktable:	
Table parallel with wheel spindle.	0.0008 inch per ft.
Transverse movement of table parallel with table surface.	0.0012 inch per ft.
Rise and fall of table in its long travel.	0.0012 inch per ft.
Universal head and tailstock:	
Taper of workhead spindle runs true, 4 inches from nose.	0.0004 inch TIR
Tailstock spindle parallel with table.	0.0012 inch per ft.
Tailstock spindle parallel with center T-slot of table.	0.0012 inch per ft.
Tailstock axis in line with workhead axis in vertical plane (workhead previously to be set parallel with table and T-slot by means of mandrel).	0.0008 inch TIR
Ditto, in horizontal plane.	0.0008 inch TIR
Wheel spindle:	
Wheel spindle for true running.	0.0004 inch TIR
Wheel spindle for axial slip.	0.0004 inch TIR
Vertical adjustment of wheelhead square with worktable in plane through long axis of table (for machines with vertically adjustable wheelhead).	0.0036 inch per ft.
Ditto, in plane perpendicular to said plane (for machines with vertically adjustable wheelhead).	0.0036 inch per ft.
Pad for center gauge on level with axis of wheel spindle.	0.004 inch

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TABLE VII.
Alignment tests (types III and IV)

Test to be applied	Maximum, allowable deviation
Swivel work table:	
Parallelism of the top of the swivel work table with its movement toward and away from the swivel center - in full length of travel.	0.001 inch TIR
Parallelism of the top of the swivel work table with its cross (transverse) movement - in 1-1/4 inch travel. (5/8 inch each side of swivel center).	0.001 inch TIR
Parallelism of top of swivel work table in its total swivel movement and indicating at 6 inch radius from swivel center.	0.002 inch TIR
Wheel spindle:	
Squareness of the vertical movement of the wheel spindle with the top of the swivel work table in both directions. In full length of travel.	0.002 inch TIR
Parallelism of the cross (transverse) movement of the wheel spindle with top of the swivel work table, in 4 inches.	0.001 inch TIR
Parallelism of the horizontal (axial) movement of the wheel spindle with the top of the swivel work table, in 4 inches.	0.001 inch TIR
Concentricity of the wheel spindle:	
Straight diameter.	0.002 inch TIR
Face.	0.001 inch TIR
Tracer grinding and cutting attachment:	
With proving bar mounted in spindle. Indicate at nose - concentric within.	0.0005 inch TIR
Indicate at end of 4 inch proving bar - concentric within.	0.0015 inch TIR
Indicate large and small diameter of taper on spindle nose - concentric within.	0.0005 inch TIR
Concentricity of taper hole in work spindle.	0.001 inch TIR
Test bar in taper of work spindle to run true at spindle nose.	0.001 inch TIR
10 inches from spindle nose.	0.002 inch TIR

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4.4.4 Electromagnetic interference control tests. Unless otherwise specified (see 6.2.1), when electromagnetic interference control is required under 3.15, the First Article shall be tested for compliance with 3.15 in accordance with table II of MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A, B, or C, as specified (see 6.2.1).

5.1.1 Level A. The machine shall be preserved and packaged in accordance with the applicable level A requirements of MIL-M-18058.

5.1.2 Level B. The machine shall be preserved and packaged as specified in 5.1.1.

5.1.3 Level C. The machine shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice, when such meets the requirements of this level.

5.1.4 Technical publications. Technical publications for each grinding machine shall be preserved in accordance with MIL-P-116, method IC-1.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. The machine shall be packed in accordance with the applicable level A requirements of MIL-M-18058.

5.2.2 Level B. The machine shall be packed in accordance with the applicable level B requirements of MIL-M-18058.

5.2.3 Level C. The machine shall be packed in a manner which will insure carrier acceptance and safe delivery at destination. Shipping containers or the method of shipment shall conform to the carrier's rules or regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order (see 6.2.1), interior packages and exterior shipping containers shall be marked in accordance with Federal Standard No. 123.

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5.3.2 Military agencies. In addition to any special marking required by the contract or order (see 6.2.1), interior packages and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Types 0, I, and II. The tool and cutter grinders covered by this specification are intended for resharpening conventional types of cutters within the range specified, including all types of milling cutters, reamers, counterbores, hobs, taps, form tools, thread chasers, etc., or other metal-cutting tools. With the proper attachments, they can be used for internal and external cylindrical grinding and surface grinding.

6.1.2 Types III and IV. The tool, cutter, gashing, flute and radius grinders covered by this specification are intended for resharpening conventional-type end milling cutters having cylindrical or tapered sides, straight or helical flutes, and ball, radius, or square nose. With proper attachments, it can be used for grinding and sizing tracer points as used on tracer-controlled milling machines as well as shell or ball end mills, face-milling cutters and form-milling cutters that can be mounted on arbors and within the capacity of the machine.

6.2 Ordering data. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type, style, and size required (see 1.2.1).
(c) First Article approval, if required (see 3.2).
(d) Operating controls, if different (see 3.3.3).
(e) Specify exceptions, and additional requirements for safety and health (see 3.3.4 and 6.5).
(f) Painting, if different (see 3.3.12).
(g) Whether bench type or pedestal type base or unit type construction is required for type 0 machines (see 3.4.2).
(h) Specify when mechanical crossfeed is required for type 0 machines (see 3.4.3).
(i) Specify when hydraulic feed is required for type 0 machines (see 3.4.4).

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(j) Specify whether single or multi-speed wheelhead (see 3.4.6).
(k) Specify if wheelhead tilt is required (see 3.4.6, 3.5.4, and 3.6.5).
(l) Universal workhead, if not required (see 3.4.7, 3.5.8, and 3.6.11).
(m) Standard equipment, if different (see 3.4.8, 3.5.9, 3.6.12, 3.7.5, 3.8.5, and 3.8.6).
(n) Grinding wheels, number of each size, type, dimensional characteristics, and material composition (see 3.4.8(f), 3.5.9(f), 3.6.12(f), 3.7.5(b), 3.8.5(d), and 3.8.6(g)).
(o) Reducing collets, as required (see 3.4.8(h), 3.5.9(h), 3.6.12(h), 3.7.4, and 3.8.4.1).
(p) Type of ways required for type I machines (see 3.5.3).
(q) Wheel speed changes, if different (see 3.5.4).
(r) Specify tailstock taper or other internal configuration required for type II machines (see 3.6.8).
(s) Chuck size, spindle taper, and motor horsepower required for cylindrical grinding attachment for type II machines (see 3.6.12(k)).
(t) The number of quill sizes and the number of grinding wheels, their shank size, type, dimensional characteristics, and material composition required for type II machines (see 3.6.12(l)).
(u) Cutter bushings, number and size required for type II machines (see 3.6.12(n)).
(v) Machine requirements, if different (see tables I, II, and III).
(w) Optional horsepower, if required (see 3.7.1 and table II).
(x) Specify when tracer grinding and turning attachment is not required (see 3.7.4 and 3.8.4.1).
(y) The way ball bearing workholding spindle for type IV machines shall be bored (see 3.8.5(f)).
(z) Size workholding taper required for type IV machines (see 3.8.6(f)).
(aa) Electrical equipment, if different (see 3.11).
(bb) Motors, if different, and voltage characteristics and voltage for initial wiring of machine, if different (see 3.11.1).
(cc) Type of enclosure for motor, if different (see 3.11.1).
(dd) Specify whether a coolant system shall be provided, and type or types acceptable (see 3.12).
(ee) Optional equipment, if required (see 3.14).
(ff) Electromagnetic interference control, when required (see 3.15).
(gg) Type fungus proofing, if required (see 3.16).
(hh) Lubrication chart or plate, if different (see 3.17).
(ii) Nameplate, if different (see 3.18.2).
(jj) Identification plate, if required (see 3.19).
(kk) First Article inspection, if different (see 4.2).
(ll) Additional inspection, if different (see 4.3).
(mm) Acceptance tests required, if different (see 4.4).
(nn) Electromagnetic interference control tests, if different (see 4.4.4 and 6.4).

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(oo) Level of preservation-packaging (see 5.1). Level of packing (see 5.2).
(pp) Special marking, when required (see 5.3.1 and 5.3.2).

6.2.2 Contract data requirements. For military procurements, required technical data, such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.19).

6.3 First Article. When First Article approval is required under 3.2, the solicitation should include complete information concerning examination, testing, and approval. For military procurements, the solicitation should adhere to the applicable portions of Section 1, Part 19, of the Armed Services Procurement Regulation.

6.4 Army and Air Force procurements. When equipment purchased for the Departments of the Army and Air Force requires electromagnetic interference control testing, such testing should be in accordance with MIL-STD-461, table II, class IIB equipment, except only Test Methods CE03 and RE02 should be required.

6.5 Occupational Safety and Health Act (OSHA). Paragraph 3.3.4 requires compliance only with OSHA requirements concerning the machine itself. These requirements include, but are not limited to, the following paragraphs of the OSHA Standards:

1910.144 - Safety color code for marking physical hazards

1910.212 - General requirements for all machines

1910.215 - Abrasive wheel machinery

1910.219 - Mechanical power transmission apparatus

Since this specification does not require compliance with OSHA requirements concerning the operating environment, procuring agencies should insert additional requirements under 6.2.1(e) that will integrate the new machine into the work environment without exceeding OSHA hazard levels. OSHA paragraphs applicable to this specification which are pertinent to the operating environment are:

1910.93 Air contaminants

1910.95 Occupational noise exposure

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The above paragraphs are cited to assist procuring agencies in complying with OSHA. The machine user should measure and evaluate his safety and health requirements based on the entire contents of OSHA standards.

6.6 Size identification. Machines covered by this specification are sized in accordance with swing diameter, inches, and maximum distance between tailstock centers, inches. When specifying optional or other sizes, the applicable table (I, II, or III) should be used as a guide.

6.7 Squirrel cage induction motors. Procuring activities should exercise due caution in requiring this motor type (see 3.11.1 and 6.2.1(bb)), particularly when variable speed motors are involved. The requirements may eliminate the commercial equipment of some, or all, potential suppliers.

6.8 Plain and universal. The terms "plain" and "universal" have been eliminated from this specification. All machines covered are considered standard models. Types 0 and I machines may be converted to the old term "universal" by the addition of cylindrical, surface, internal, gear cutter sharpening attachments.

6.9 Type III. In a previous revision, type III machine was deleted because it was no longer commercially available. The type III machine in this revision is a new and distinct machine type.

MILITARY CUSTODIANS:

Army - WC
Navy - SH
Air Force - 84

Review activities:

Army - WC, ME
Navy - SH, YD
Air Force - 84
DSA - GS

User activities:

Army - SM, MU
Navy - MC

Preparing activity:

DSA-IP

Project number:

3415-0101

CIVIL AGENCIES COORDINATING ACTIVITY:

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DOCUMENT IDENTIFIER AND TITLE: OO-G-692C - Grinding Machines, Tool and Cutter; and Grinding and Gashing Machines, Tool, Cutter, Flute and Radius

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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